SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Poster Presentation

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4- Scenario based**	**Assessment:**	**Assessment Tool 1 - Scenario based**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	V.Naga Nischay	**Reg. No.:**	192472255
Section / Batch:	CSE(AI)	**Date:**	20-08-2026

Code of Conduct

I V.Naga Nischay (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nischay

Criteria	Weigh t age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understandin g of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding ; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution	20	Innovative,	Logical	Basic solutions	Incorrect or no	

Design		**feasible solutions with strong justification**	**solutions with adequate justification**	**with weak justification**	**solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

1. Higher Education — Hypothesis Testing Using CI

A university claims that at least 80% of its students are satisfied with online classes. A survey produces a 95% confidence interval of (74%, 79%) for the satisfaction proportion. What conclusion can be made about the university's claim?

Solution

1. Population and Sample: Population: all students of the university enrolled in online classes. Sample: the surveyed group of students used to construct the interval.

2. Parameter of Interest: The true population proportion of students satisfied with online classes, p.

3. Point Estimate: The point estimate lies at the centre of the given interval, i.e. approximately (74% + 79%)/2 = 76.5%.

4. Null Hypothesis (H0): $p \geq 0.80$ (the university's claim — at least 80% of students are satisfied).

5. Alternative Hypothesis (H1): $p < 0.80$ (fewer than 80% of students are satisfied).

6. Significance Level (α): $\alpha = 0.05$ (95% confidence, matching the given interval).

7. Confidence Interval: The given 95% CI for the satisfaction proportion is (74%, 79%).

8. Interpretation of the Interval: The claimed value of 80% does not fall inside the interval (74%, 79%); in fact the entire interval lies below 80%.

9. Compare with Claimed Value: Since the upper bound of the CI (79%) is still less than the claimed 80%, the data is inconsistent with the university's claim at the 95% confidence level.

10. Industry-Oriented Conclusion: There is sufficient statistical evidence to reject the university's claim. The true satisfaction rate is likely between 74% and 79%, significantly below the claimed 80%. The university should not publicize the '80% satisfaction' figure and should instead investigate the drivers of dissatisfaction with online classes (e.g., engagement, instructor interaction, technical issues) to improve the student experience.

2. Mobile Application Development — Hypothesis Testing Using CI

A mobile application developer claims that users spend an average of 30 minutes per day on the application. A sample produces a 99% confidence interval of (31.5, 35.2) minutes. Using the confidence interval, determine whether the claim is supported.

Solution

1. Population and Sample: Population: all users of the mobile application. Sample: the group of users whose daily usage times were recorded to build the interval.

2. Parameter of Interest: The true population mean daily usage time, μ (in minutes).

3. Point Estimate: The point estimate lies at the centre of the interval, i.e. approximately $(31.5 + 35.2)/2 = 33.35$ minutes.

4. Null Hypothesis (H_0): $\mu = 30$ minutes (the developer's claimed average usage time).

5. Alternative Hypothesis (H_1): $\mu \neq 30$ minutes (the true average usage time differs from the claim).

6. Significance Level (α): $\alpha = 0.01$ (99% confidence, matching the given interval).

7. Confidence Interval: The given 99% CI for average daily usage is (31.5, 35.2) minutes.

8. Interpretation of the Interval: The claimed value of 30 minutes does not fall inside the interval (31.5, 35.2); the entire interval lies above 30 minutes.

9. Compare with Claimed Value: Since even the lower bound of the CI (31.5) exceeds 30, the data is inconsistent with the claim at the 99% confidence level.

10. Industry-Oriented Conclusion: The claim of a 30-minute average usage time is not supported by the data — users actually spend significantly more time on the app, with the true average likely between 31.5 and 35.2 minutes. This is favourable for the developer from an engagement standpoint, but the publicly stated '30 minutes' figure understates actual usage and should be updated in product/marketing materials and investor reporting.

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3. Financial Technology / Fraud Detection — Hypothesis Testing Using P-value

A credit-card company claims that its fraud detection system identifies 95% of fraudulent transactions. After testing the system on a sample, the calculated p-value is 0.02. At a 5% significance level, what conclusion should the data scientist make?

Solution

1. Population and Sample: Population: all fraudulent transactions the system could encounter. Sample: the transactions used to test the fraud detection system's identification rate.

2. Parameter of Interest: The true population proportion of fraudulent transactions correctly identified by the system, p.

3. Point Estimate: Not given directly, but the test was performed against the claimed detection rate of 95%.

4. Null Hypothesis (H0): $p = 0.95$ (the system detects fraud at the claimed rate of 95%).

5. Alternative Hypothesis (H1): $p \neq 0.95$ (the system's true detection rate differs from the claimed 95%).

6. Significance Level (α): $\alpha = 0.05$.

7. Test Statistic: Not provided explicitly, but it corresponds to the reported p-value below.

8. P-value: Given p-value = 0.02.

9. Compare p-value with α: p-value (0.02) < α (0.05), so the result is statistically significant.

10. Industry-Oriented Conclusion: There is sufficient evidence to reject H0. The fraud detection system's actual performance differs significantly from the claimed 95% detection rate. The data scientist should flag this discrepancy to stakeholders, investigate whether the system is under- or over-performing relative to the claim, and recommend recalibration or further validation before relying on the stated 95% figure in compliance or marketing communications.

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4. E-commerce — Hypothesis Testing Using P-value

An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The *hypothesis test produces a p-value of 0.03. At α = 0.05, determine whether the new algorithm significantly improves conversion.*

Solution

1. Population and Sample: Population: all customer visits/orders processed through the new recommendation algorithm. Sample: the visits observed after implementation, yielding a 15% conversion rate.

2. Parameter of Interest: The true population conversion proportion, p, under the new recommendation algorithm.

3. Point Estimate: $\hat{p} = 0.15$ (15% observed conversion rate).

4. Null Hypothesis (H0): $p = 0.12$ (the new algorithm performs the same as the existing 12% baseline).

5. Alternative Hypothesis (H1): $p > 0.12$ (the new algorithm improves the conversion rate above the baseline — one-tailed test).

6. Significance Level (α): $\alpha = 0.05$.

7. Test Statistic: Not explicitly given, but consistent with the reported p-value below.

8. P-value: Given p-value = 0.03.

9. Compare p-value with α: p-value (0.03) < α (0.05), so the result is statistically significant.

10. Industry-Oriented Conclusion: There is sufficient evidence to reject H0. The new recommendation algorithm significantly improves the conversion rate compared to the existing 12% baseline (observed 15%). The e-commerce company can confidently roll out the new algorithm platform-wide, as the improvement is unlikely to be due to random chance.

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5. Healthcare — Hypothesis Testing Using P-value

A hospital introduces an AI-based patient scheduling system. The hospital wants to determine whether it reduces average waiting time. A hypothesis test produces a p-value of 0.08. Using a significance level of 5%, should the hospital conclude that the new system reduces waiting time?

Solution

1. Population and Sample: Population: all patients scheduled through the hospital's system. Sample: the patients whose waiting times were recorded after the AI-based scheduling system was introduced.

2. Parameter of Interest: The true population mean patient waiting time, μ, under the new AI-based scheduling system.

3. Point Estimate: Not given directly, but reflected in the test statistic underlying the reported p-value.

4. Null Hypothesis (H0): $\mu = \mu0$ (the new AI-based scheduling system does not reduce average waiting time compared to the previous system).

5. Alternative Hypothesis (H1): $\mu < \mu0$ (the new system reduces average waiting time — one-tailed test).

6. Significance Level (α): $\alpha = 0.05$.

7. Test Statistic: Not explicitly given, but consistent with the reported p-value below.

8. P-value: Given p-value = 0.08.

9. Compare p-value with α: p-value (0.08) > α (0.05), so the result is not statistically significant.

10. Industry-Oriented Conclusion: There is insufficient evidence to reject H0. Based on this test, the hospital cannot conclude that the AI-based scheduling system significantly reduces average patient waiting time at the 5% significance level, although the p-value (0.08) is fairly close to the threshold, suggesting a possible mild effect. The hospital should consider collecting more data (larger sample size) or running the system longer before making a final rollout decision, as the current evidence is suggestive but not conclusive.